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Felix: a scalable natural proof assistant

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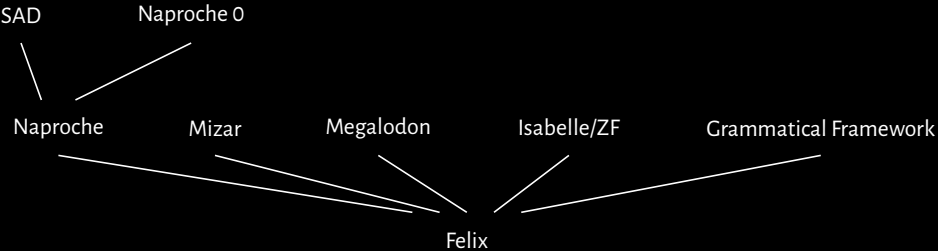
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Example: Cantor's theorem in Felix

Theorem (Cantor). There exists no surjection from A to 2^A .

Proof. Suppose not. Consider a surjection f from A to 2^A . Let $B = \{a \in A \mid a \notin f(a)\}$. Then $B \in 2^A$. There exists $a' \in A$ such that $f(a') = B$. Now $a' \in B$ iff $a' \notin f(a') = B$. Contradiction.

Genealogy



Language improvements over Naproche

Smart paradigms

Proper number agreement

No variable vs. word ambiguities

Extensible declarative grammar

Helpful parser error messages

Structures

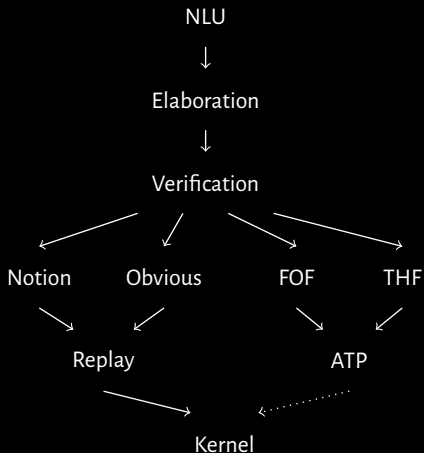
Inductive definitions

Relations can take parameters

More expressive calculations

Comprehensions as proper terms

Architecture



Theory graph generation, tokenizing, lexical extraction, adaptive grammar, Earley-parsing

Name resolution, typing, formation checking

Declaration checking,
obligation generation

Bounded internal reasoning
or export of obligation

Internal reasoning builds proof objects
No reconstruction for external ATPs yet

Higher-order Tarski–Grothendieck
Monomorphic HOL

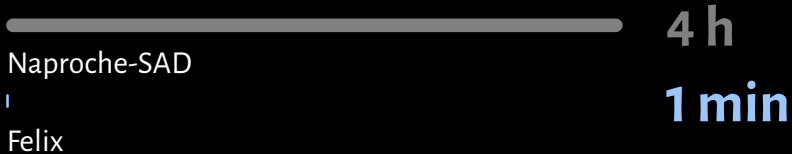
Scale

Relative size of Naproche formalization corpora in approximate SLOC

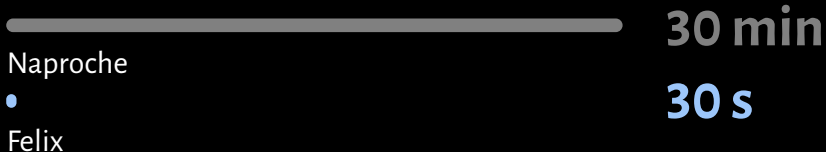
Ramsey 540	Linear algebra 2k
Geometry 2k	
10 of 100 theorems 2.5k	
Perfectoid rings 3.7k	
Representation theory 4.2k	
Real analysis 5.4k	
Set theory 13.8k	

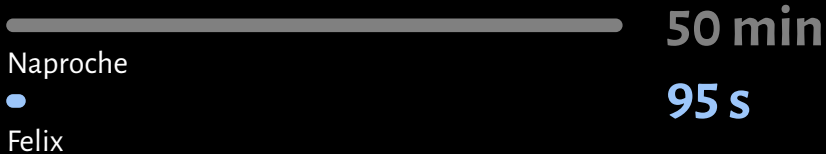
Felix's library
206k

Silver's theorem on singular cardinals



Hausdorff's theorem on regularity of successor cardinals





Why is Felix faster?

Faster lexer, parser, &c.

One attempt per obligation

Theory-level parallelism

Streaming obligation generation

ATP worker pool with speculative execution

Incremental architecture

More explicit citations → smaller ATP problems

Obvious inferences

Less class encoding compared to KM

Notion checking without ATP

Autoformalization

Basic Group Theory	Stone Representation Theorem	Cardinal Arithmetic	Continuous Extensions	
Real Completeness	Boolean Algebras	Transfinite Recursion	Lattice Theory	Urysohn's Lemma
Clopen Sets	Solovay Stationary Splitting	Ultrafilters	Isbell–Mrówka Space	Surreal Numbers
Hartogs's Lemma	Order Theory	Set Theory	Stationary Sets	Erdős–Dushnik–Miller Theorem
Ordinal Theory	Infinite Combinatorics	Burali–Forti Paradox	General Topology	Zorn's Lemma
Fixed-Point Theory	Stone Spaces	Compactness	Lagrange's Theorem	Uniform Limit Theorem
Compact Hausdorff Spaces	Least-Upper-Bound Property	Stationary Partitions	Fodor's Lemma	
Hausdorff Maximal Principle	Stone Space Compactness	Cayley's Theorem	Transfinite Induction	
Cantor's Theorem	Tietze Extension Theorem	Separation Axioms	Finite Birkhoff Representation	
Silver's Theorem	Well-Founded Recursion	Knaster–Tarski Theorem	Infinite Cardinal Arithmetic	

Tietze extension theorem

Theorem (Tietze extension)

Let X be a normal space. Let A be closed in X . Suppose $f \in A \rightarrow \mathbb{R}$ and f is continuous from $X \upharpoonright A$ to $\mathbb{R}$. Then there exists $F \in X \rightarrow \mathbb{R}$ such that F is continuous from X to $\mathbb{R}$ and $F \upharpoonright A = f$.

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\begin{theorem}[Tietze extension]\label{tietze_extension_real}
  Let  $X$  be a normal space.
  Let  $A$  be closed in  $X$ .
  Suppose  $f \in \text{funcs}\{A\}\{\mathbb{R}\}$  and  $f$  is continuous from
     $\text{subspaceTopology}\{X\}\{A\}$  to  $\mathbb{R}$ .
  Then there exists  $F \in \text{funcs}\{\text{carrier}[X]\}\{\mathbb{R}\}$  such that
     $F$  is continuous from  $X$  to  $\mathbb{R}$  and
     $\text{restr}\{F\}\{A\}=f$ .
\end{theorem}
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Solovay stationary splitting

Corollary

Let κ be a regular cardinal. Suppose $\omega \in \kappa$ and S is stationary in κ . Then there exists P such that $P \in \kappa \rightarrow 2^S$ and $\bigcup \text{ran}(P) = S$ and for every $r \in \kappa$ we have $P(r)$ is stationary in κ and for all $i, j \in \kappa$ such that $i \neq j$ we have $P(i)$ is disjoint from $P(j)$.

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\begin{corollary}\label{solovay_stationary_splitting_expanded}
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  Let  $\kappa$  be a regular cardinal.
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  Suppose  $\omega \in \kappa$ .
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```
  Suppose  $S$  is stationary in  $\kappa$ .
```

```
  Then there exists  $P$  such that
```

```
     $P \in \kappa \rightarrow 2^S$  and
```

```
     $\bigcup \text{ran}(P) = S$  and
```

```
    for every  $r \in \kappa$  we have
```

```
       $P(r)$  is stationary in  $\kappa$  and
```

```
    for all  $i, j \in \kappa$  such that  $i \neq j$  we have
```

```
       $P(i)$  is disjoint from  $P(j)$ .
```

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\end{corollary}
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More to come...